

Quantum Mechanics

Lecture 6: Linear and Circular Polarization, Operators, and Expectation Values

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Chapter 1

Linear and Circular Polarization, Operators, and Expectation Values

1.1 The Smallest Useful Quantum System

Take a photon moving in the z -direction with definite momentum and set aside its translational motion. What remains is its transverse polarization. Classically we picture an electric field oscillating along an axis in the xy -plane. For a single photon that transverse degree of freedom becomes a particularly clean quantum system: the state space is only two-dimensional, but it already contains superposition, incompatible measurements, complex amplitudes, and probabilistic outcomes.

Begin with an ideal linear polarizer. The conducting direction of a wire-grid polarizer supports a current and therefore suppresses the electric field component parallel to the wires. The perpendicular component is transmitted. Consequently the *transmission axis*, rather than the wire direction, labels the polarizer.

The same apparatus has two roles. If a photon emerges from a polarizer, the apparatus has prepared a state. If a prepared photon is sent into a second polarizer, that apparatus tests the state. Equal preparation and test axes give certain transmission; orthogonal axes give certain rejection.

Question from the audience. Could ordinary classical particles imitate this first experiment?^a

Response. Yes. One could imagine bullets carrying little flags and a mechanical slit that passes one flag orientation while blocking the other. Nothing nonclassical has appeared while the only preparations and tests are the same two orthogonal directions.

^aLecture timestamp: 00:09:01–00:09:25.

The difficulty begins with rotational symmetry. A polarizer can be turned to any angle, so a photon can be prepared along a continuum of linear polarization axes. A fixed analyzer nevertheless has only two outcomes: the photon passes or it does not. No single photon divides into a transmitted fraction and a rejected fraction. Continuous preparations and discrete outcomes must coexist.

This is the central puzzle. A classical wave divides continuously, and the flagged-bullet model remains deterministic. A photon at an intermediate angle instead gives one indivisible event whose probability depends on the relative orientation of the apparatus.

Mathematically this polarization problem is isomorphic to a spin-one-half system: both use a

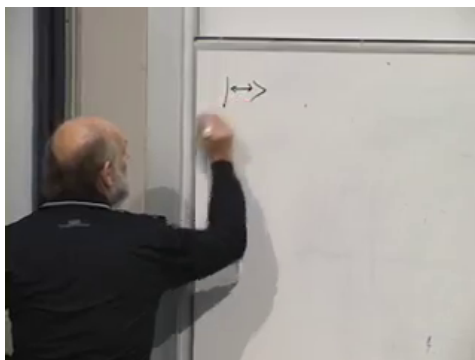


Figure 1.1: The double-headed mark records a polarization axis without assigning it a preferred direction.

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two-dimensional complex state space and noncommuting two-outcome observables. Polarization axes are somewhat easier to picture, so they provide the cleaner geometry for the present discussion.

1.2 The Axial Basis and Its Observable

Choose horizontal and vertical transmission axes and call them x and y . Linear polarization is axial: horizontal-to-the-right is not different from horizontal-to-the-left. Reversing the drawn arrow leaves the same polarization axis.

Represent the two orthogonal states by

$$|x\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad |y\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}. \quad (1.1)$$

They obey

$$\langle x|x\rangle = \langle y|y\rangle = 1, \quad \langle x|y\rangle = 0. \quad (1.2)$$

These are two basis vectors, not the only two state vectors. Every normalized superposition

$$|\psi\rangle = a|x\rangle + b|y\rangle, \quad |a|^2 + |b|^2 = 1, \quad (1.3)$$

is a possible pure polarization state. Multiplication of the entire vector by one common phase does not change the physical state.

Question from the audience. Are the entries 1, 0 and 0, 1 completely arbitrary?^a

Response. The coordinate convention is chosen with the axes, but the vectors must be normalized and mutually orthogonal. Those conditions encode total probability one and perfect distinguishability. Their common overall phase is irrelevant, so several column vectors can represent the same physical ray.

^aLecture timestamp: 00:16:50–00:18:38.

To turn a test into an observable, attach numbers to its outcomes. Record +1 when an x -analyzer transmits the photon and -1 when it rejects it. The labels are conventional; 7 and 12 would distinguish the outcomes too. The symmetric choice ± 1 gives

$$\hat{P}_0|x\rangle = +|x\rangle, \quad \hat{P}_0|y\rangle = -|y\rangle, \quad (1.4)$$

Figure 1.2: Orthogonality and the first matrix action in the $(|x\rangle, |y\rangle)$ basis.

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and therefore

$$\hat{P}_0 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = |x\rangle\langle x| - |y\rangle\langle y|. \quad (1.5)$$

The operator can be constructed from known eigenstates and recorded values, as we have just done. In other problems the order is reversed: one begins with an operator and solves for its eigenvectors and eigenvalues. For example, the wave-number operator $-i d/dx$ is given first, and its plane-wave eigenstates are then found. Either route expresses the same postulate: an observable has definite value λ in a state satisfying $\hat{A}|\lambda\rangle = \lambda|\lambda\rangle$.

1.3 A Second Basis at Forty-Five Degrees

Turn the preparation polarizer through 45° . The transmitted state has equal x - and y -components:

$$|d_+\rangle = \frac{|x\rangle + |y\rangle}{\sqrt{2}} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}. \quad (1.6)$$

Its orthogonal companion, the backslash polarization, can be chosen as

$$|d_-\rangle = \frac{|x\rangle - |y\rangle}{\sqrt{2}} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix}. \quad (1.7)$$

Putting the minus sign on the other component multiplies the whole state by -1 and changes no physics. Directly,

$$\langle d_- | d_+ \rangle = \frac{1}{2}(1 - 1) = 0. \quad (1.8)$$

The amplitude that a slash-polarized photon passes an x -analyzer is

$$\langle x | d_+ \rangle = \frac{1}{\sqrt{2}}. \quad (1.9)$$

The probability is the absolute square,

$$\text{Prob}(x|d_+) = |\langle x | d_+ \rangle|^2 = \frac{1}{2}, \quad \text{Prob}(y|d_+) = \frac{1}{2}. \quad (1.10)$$

The amplitude and the probability are not interchangeable. Amplitudes may be negative or complex; probabilities are nonnegative real numbers.

Now attach +1 to the slash outcome and -1 to the backslash outcome. The corresponding observable is

$$\hat{P}_{\pi/4} = |d_+\rangle \langle d_+| - |d_-\rangle \langle d_-| = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}. \quad (1.11)$$

Exchanging which diagonal outcome is called +1 simply replaces this operator by $-\hat{P}_{\pi/4}$. Indeed,

$$\hat{P}_{\pi/4} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad \hat{P}_{\pi/4} \begin{pmatrix} 1 \\ -1 \end{pmatrix} = - \begin{pmatrix} 1 \\ -1 \end{pmatrix}. \quad (1.12)$$

Question from the audience. Does knowing the state determine what is observable?^a

Response. No. The state determines which observables, if any, have definite values. In $|d_+\rangle$, the diagonal observable is definite while $\hat{P}_0$ gives two outcomes with equal probability. Similarly, a position delta state makes position definite, whereas a plane wave makes momentum definite; a generic wavefunction usually makes neither definite.

^aLecture timestamp: 00:35:14–00:37:32.

1.4 Arbitrary Linear Polarization

Let the transmission axis make angle θ with the x -axis. Its state and orthogonal partner are

$$|\theta\rangle = \cos \theta |x\rangle + \sin \theta |y\rangle = \begin{pmatrix} \cos \theta \\ \sin \theta \end{pmatrix}, \quad (1.13)$$

$$|\theta_\perp\rangle = |\theta + \pi/2\rangle = -\sin \theta |x\rangle + \cos \theta |y\rangle = \begin{pmatrix} -\sin \theta \\ \cos \theta \end{pmatrix}. \quad (1.14)$$

Both have unit norm, and

$$\langle \theta | \theta_\perp \rangle = -\cos \theta \sin \theta + \sin \theta \cos \theta = 0. \quad (1.15)$$

Also $|\theta + \pi\rangle = -|\theta\rangle$, confirming that a linear polarization axis has period π , not 2π .

Suppose the first polarizer prepares $|\theta\rangle$ and the second tests the x -direction. Then

$$\text{Prob}(x|\theta) = |\langle x | \theta \rangle|^2 = \cos^2 \theta, \quad \text{Prob}(y|\theta) = \sin^2 \theta. \quad (1.16)$$

Their sum is one. Reversing the inner product does not change the probability because $\langle \theta | x \rangle = \langle x | \theta \rangle^*$.

For two arbitrary axes α and β , the needed trigonometric identity is

$$\cos(\alpha - \beta) = \cos \alpha \cos \beta + \sin \alpha \sin \beta. \quad (1.17)$$

$$\begin{aligned} \sin\theta |y\rangle &= \begin{pmatrix} \cos\theta \\ \sin\theta \end{pmatrix} \\ + \cos\theta |y\rangle &= \begin{pmatrix} -\sin\theta \\ \cos\theta \end{pmatrix} \end{aligned}$$

Figure 1.3: The arbitrary linear-polarization state and its orthogonal companion written in the axial basis.

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$$\begin{aligned} \cos(\alpha+\beta) &= \cos\alpha \cos\beta - \sin\alpha \sin\beta \\ \cos(\alpha-\beta) &= \cos\alpha \cos\beta + \sin\alpha \sin\beta \\ \sin(\alpha+\beta) &= \sin\alpha \cos\beta + \sin\beta \cos\alpha \\ \sin(\alpha-\beta) &= \sin\alpha \cos\beta - \sin\beta \cos\alpha \end{aligned}$$

Figure 1.4: The trigonometric identities used to compare two rotated polarization bases.

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Handwritten equations on a whiteboard:

$$\langle \beta | \alpha \rangle = \cos \alpha \cos \beta + \sin \alpha \sin \beta$$

$$\cos(\alpha - \beta)$$

$$|\langle \beta | \alpha \rangle|^2 = \cos^2(\alpha - \beta)$$

Figure 1.5: The overlap is an amplitude; its absolute square is the transmission probability.

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The related sum, sine, and double-angle identities are useful enough to keep beside it.

The transition amplitude is

$$\langle \beta | \alpha \rangle = \begin{pmatrix} \cos \beta & \sin \beta \end{pmatrix} \begin{pmatrix} \cos \alpha \\ \sin \alpha \end{pmatrix} \quad (1.18)$$

$$= \cos \alpha \cos \beta + \sin \alpha \sin \beta \quad (1.19)$$

$$= \cos(\alpha - \beta). \quad (1.20)$$

Therefore the single-photon form of Malus's law is

$$\boxed{\text{Prob}(\beta | \alpha) = |\langle \beta | \alpha \rangle|^2 = \cos^2(\alpha - \beta).} \quad (1.21)$$

Only the relative angle appears, as rotational symmetry requires. At equal angles the probability is one; at orthogonal angles it is zero. For every intermediate angle each photon still gives one of two discrete records.

1.5 How an Extra Polarizer Opens a Closed Path

Prepare $|\theta\rangle$ and test it directly with the orthogonal analyzer $\theta + \pi/2$. The transmission probability vanishes:

$$\text{Prob}(\theta + \pi/2 | \theta) = \cos^2(\pi/2) = 0. \quad (1.22)$$

Now insert an x -polarizer between them. Passing the intermediate polarizer has probability $\cos^2 \theta$, and passage prepares the new state $|x\rangle$. Conditional on that event, transmission through the last

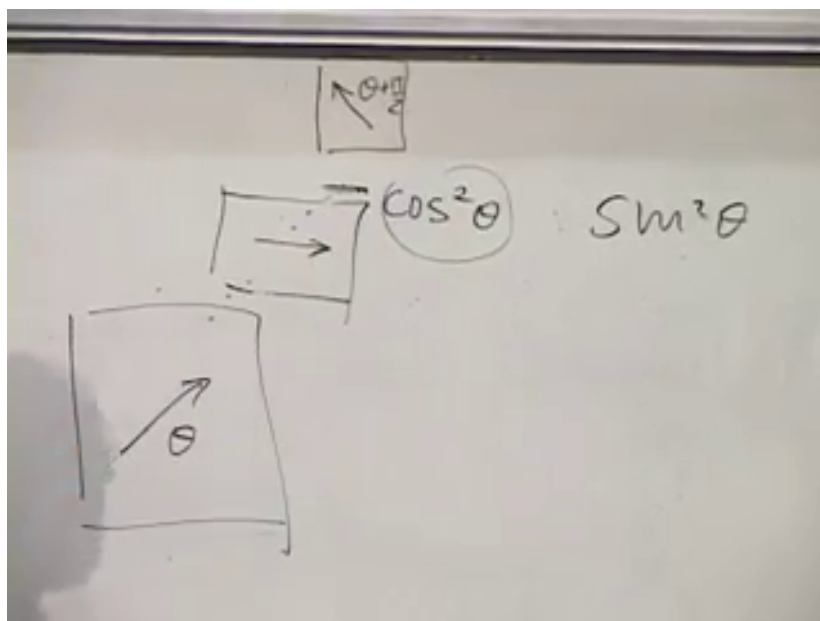


Figure 1.6: An intermediate horizontal polarizer changes the state and produces the nonzero product $\cos^2 \theta \sin^2 \theta$.

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analyzer has probability

$$\text{Prob}(\theta + \pi/2|x) = \cos^2(\theta + \pi/2) = \sin^2 \theta. \quad (1.23)$$

The probability of completing both stages is therefore

$$\text{Prob}(\theta \longrightarrow x \longrightarrow \theta + \pi/2) = \cos^2 \theta \sin^2 \theta. \quad (1.24)$$

Question from the audience. How can inserting another obstacle increase the chance of transmission?^a

Response. Without the middle apparatus the initial and final states are orthogonal. The intermediate polarizer is not a passive filter on an unchanged state: successful passage prepares $|x\rangle$, which has a nonzero overlap with the final state. The two conditional probabilities multiply. Adding a measurement can therefore open a route that was absent in the direct experiment.

^aLecture timestamp: 00:59:29–01:03:40.

Question from the audience. Why can crossed polarizers show colors when stressed transparent material is placed between them? ^a

Response. The exchange in the lecture leaves the mechanism open and notes that real optical elements can have wavelength-dependent efficiency. More specifically, stressed transparent materials are often birefringent: they give two polarization components a wavelength-dependent relative phase. The final analyzer converts that phase delay into wavelength-dependent intensity. The photon frequency is not changed by an ideal static polarizer; different frequencies are transmitted with different probabilities.

^aLecture timestamp: 01:03:42–01:05:00.

Figure 1.7: The board check of the general polarization matrix on $(\cos \theta, \sin \theta)^T$.

Lecture frame: 01:10:30.

1.6 The Observable at Angle θ

The general analyzer must have $|\theta\rangle$ and $|\theta_\perp\rangle$ as eigenvectors:

$$\hat{P}_\theta |\theta\rangle = +|\theta\rangle, \quad \hat{P}_\theta |\theta_\perp\rangle = -|\theta_\perp\rangle. \quad (1.25)$$

Its spectral form is already the answer:

$$\hat{P}_\theta = |\theta\rangle \langle \theta| - |\theta_\perp\rangle \langle \theta_\perp|. \quad (1.26)$$

In the x, y basis,

$$\hat{P}_\theta = \begin{pmatrix} \cos^2 \theta - \sin^2 \theta & 2 \sin \theta \cos \theta \\ 2 \sin \theta \cos \theta & \sin^2 \theta - \cos^2 \theta \end{pmatrix} \quad (1.27)$$

$$= \begin{pmatrix} \cos 2\theta & \sin 2\theta \\ \sin 2\theta & -\cos 2\theta \end{pmatrix}. \quad (1.28)$$

Let $c = \cos \theta$ and $s = \sin \theta$. Acting on the proposed $+1$ eigenvector gives

$$\hat{P}_\theta \begin{pmatrix} c \\ s \end{pmatrix} = \begin{pmatrix} c^2 - s^2 & 2sc \\ 2sc & s^2 - c^2 \end{pmatrix} \begin{pmatrix} c \\ s \end{pmatrix} \quad (1.29)$$

$$= \begin{pmatrix} c(c^2 + s^2) \\ s(c^2 + s^2) \end{pmatrix} = \begin{pmatrix} c \\ s \end{pmatrix}. \quad (1.30)$$

Rotating the analyzer by 90° exchanges the two outcomes:

$$\hat{P}_{\theta+\pi/2} = -\hat{P}_\theta. \quad (1.31)$$

The trigonometric reason is that both $\cos(2\theta + \pi)$ and $\sin(2\theta + \pi)$ acquire a minus sign. The physical reason is simpler: the state formerly labelled $+1$ is now labelled -1 , and conversely.

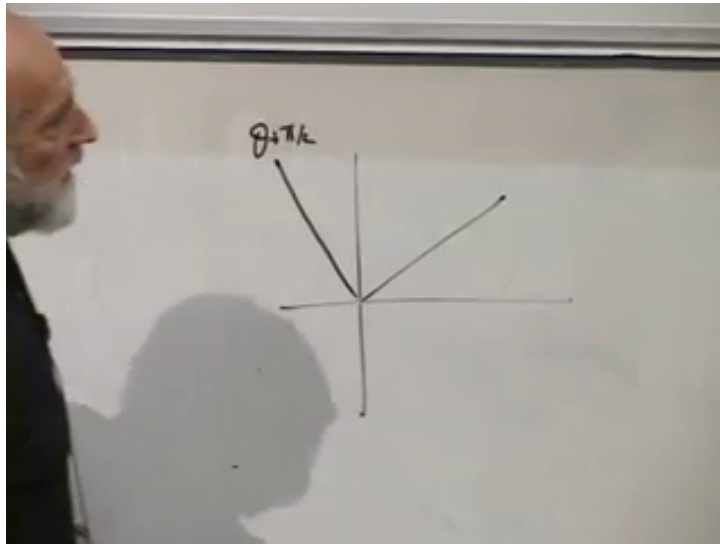


Figure 1.8: The two orthogonal analyzer axes whose outcome labels are exchanged by a 90° rotation.

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1.7 Complex Amplitudes and Circular Polarization

Real coefficients describe every linear polarization, but not every vector in a complex two-dimensional Hilbert space. Consider

$$|c_+\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ i \end{pmatrix}, \quad |c_-\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -i \end{pmatrix}. \quad (1.32)$$

Complex conjugation is essential:

$$\langle c_+ | c_+ \rangle = \frac{1}{2} (1 + (-i)i) = 1, \quad (1.33)$$

and

$$\langle c_- | c_+ \rangle = \frac{1}{2} (1 + i^2) = 0. \quad (1.34)$$

The tempting product without conjugating the bra gives the wrong answer.

To see their wave meaning, restore a dimensionally explicit phase $\phi = kz - \omega t$, with $\omega = ck$. Equal in-phase components,

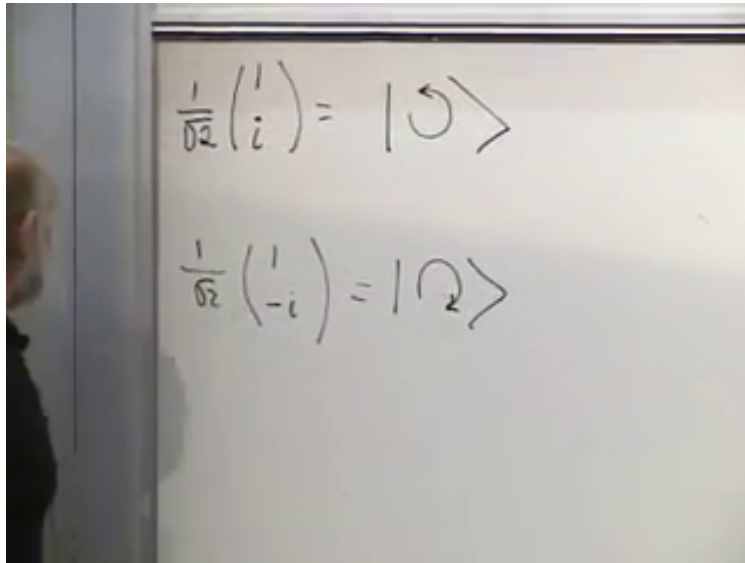
$$E_x = E_0 \cos \phi, \quad E_y = E_0 \cos \phi, \quad (1.35)$$

make a linearly polarized wave at 45° . A relative quarter-cycle shift gives

$$E_x = E_0 \sin \phi, \quad E_y = E_0 \cos \phi. \quad (1.36)$$

At a fixed point in space, the tip of $\mathbf{E} = E_x \hat{\mathbf{x}} + E_y \hat{\mathbf{y}}$ moves around a circle. Reversing the sign of one quadrature reverses the sense of rotation. The magnetic field is transverse as well; in vacuum, $\mathbf{B} = \hat{\mathbf{z}} \times \mathbf{E}/c$.

Which state is called right- or left-circular depends on the propagation, time, and viewing conventions. The invariant content here is the orthogonal pair $(1, \pm i)^T / \sqrt{2}$.



A whiteboard with two equations written in black marker. The first equation is $\frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ i \end{pmatrix} = | \uparrow \rangle$ and the second equation is $\frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -i \end{pmatrix} = | \downarrow \rangle$. The vectors are represented as column matrices.

Figure 1.9: The two normalized complex vectors that form the circular polarization basis.

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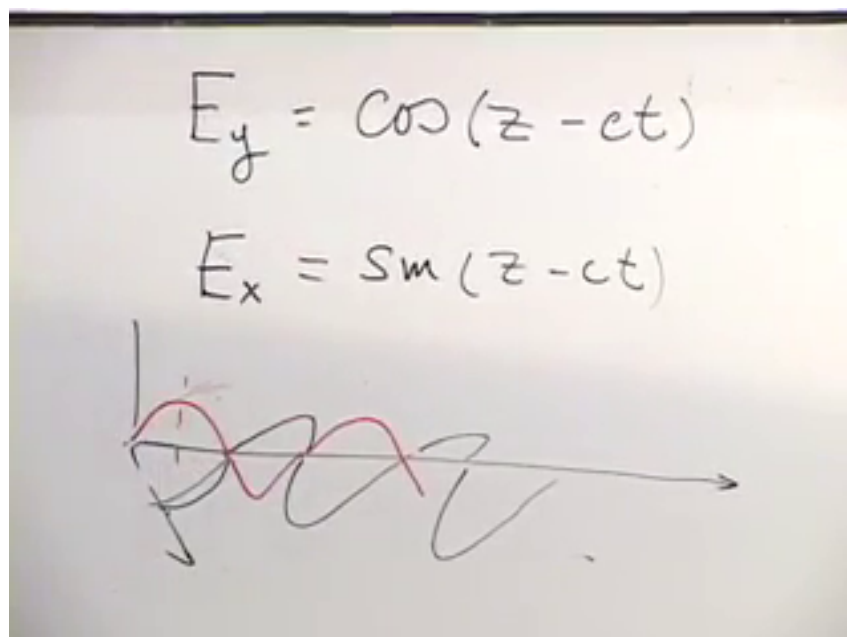


Figure 1.10: The x - and y -components are displaced by one quarter cycle.

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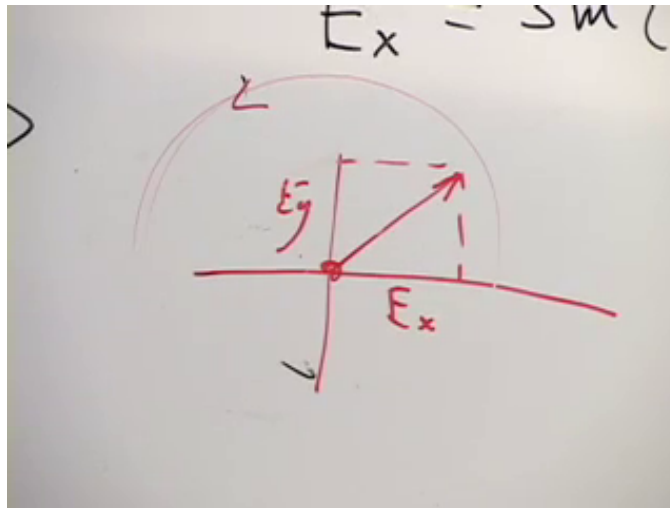


Figure 1.11: At fixed z , the two quadratures make the electric-field vector rotate.

Lecture frame: 01:25:30.

Question from the audience. What optical device creates the relative phase required for circular polarization?^a

Response. A quarter-wave plate. Its two principal axes accumulate different phases; a linear state with equal components along those axes emerges with a relative phase of $\pi/2$, producing circular polarization. Unequal component magnitudes generally produce an ellipse.

^aLecture timestamp: 01:26:37–01:27:05.

For any linear analyzer,

$$\langle \theta | c_+ \rangle = \frac{1}{\sqrt{2}}(\cos \theta + i \sin \theta). \quad (1.37)$$

Hence

$$|\langle \theta | c_+ \rangle|^2 = \frac{1}{2}(\cos \theta + i \sin \theta)(\cos \theta - i \sin \theta) \quad (1.38)$$

$$= \frac{1}{2}(\cos^2 \theta + \sin^2 \theta) = \frac{1}{2}. \quad (1.39)$$

A circularly polarized photon therefore passes every linear analyzer with probability one half. Its statistics reveal no preferred linear axis.

The general normalized pair $a, b \in \mathbb{C}$, modulo a common phase, describes elliptical polarization. A real relative phase gives linear polarization. Equal magnitudes with relative phase $\pm\pi/2$ give the two circular polarizations. Intermediate magnitudes and phases give ellipses.

$$(c + is)(c - is) = \frac{1}{2}(c^2 + s^2)$$

$$\langle \theta | 5 \rangle$$

Figure 1.12: The complex-conjugate product removes the phase and leaves transmission probability $1/2$.

Lecture frame: 01:29:50.

Question from the audience. What observable distinguishes the two circular states? ^a

Response. The lecture leaves this construction as an exercise. With $|c_+\rangle$ assigned eigenvalue $+1$ and $|c_-\rangle$ assigned -1 , the spectral rule gives

$$\hat{P}_c = |c_+\rangle \langle c_+| - |c_-\rangle \langle c_-| = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}.$$

Changing the handedness convention or exchanging the labels multiplies the operator by -1 .

^aLecture timestamp: 01:32:19–01:32:36.

1.8 Compatible and Incompatible Polarization Tests

The axial, diagonal, arbitrary linear, and circular observables do not usually share eigenvectors. In the present two-state space,

$$\hat{P}_\theta = \cos 2\theta \sigma_z + \sin 2\theta \sigma_x, \quad (1.40)$$

so two linear analyzers satisfy

$$[\hat{P}_\alpha, \hat{P}_\beta] = 2i \sin(2(\beta - \alpha)) \sigma_y. \quad (1.41)$$

They commute only when their axes are the same or orthogonal, modulo the π -periodicity of a linear polarization axis. In the orthogonal case $\hat{P}_{\theta+\pi/2} = -\hat{P}_\theta$, so measuring one immediately fixes the other with the opposite label.

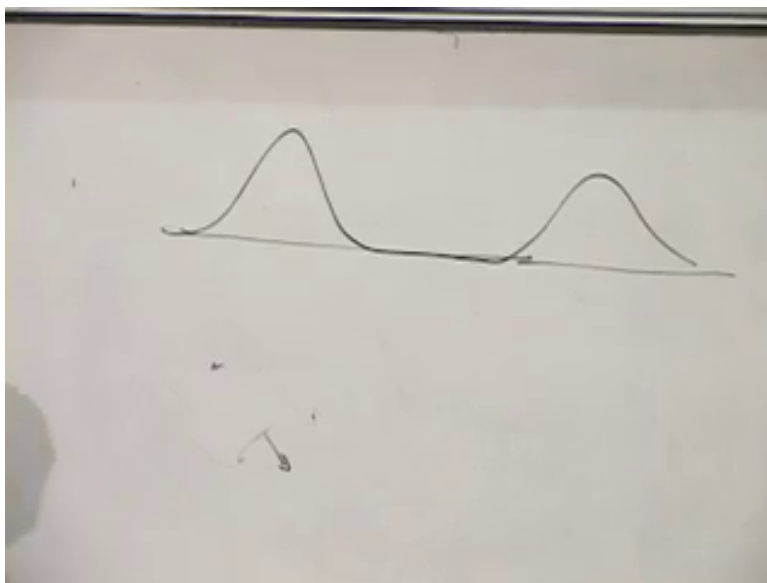


Figure 1.13: A symmetric two-lobe profile can have average position zero even though its central probability is very small.

Lecture frame: 01:36:09.

Question from the audience. Can x - and y -polarization be measured simultaneously, and what changes at 45° ? ^a

Response. The x and y outcome labels belong to one common basis and are negatives of one another in the ± 1 convention. Knowing one fixes the other. The x - and 45° -analyzers are genuinely different: their operators do not commute and have no common eigenstate. The same incompatibility holds between a linear analyzer and the circular observable.

^aLecture timestamp: 01:33:31–01:35:20.

Incompatibility is therefore not vagueness about an apparatus. Each apparatus is perfectly definite. It is the impossibility of one state being a simultaneous eigenstate of two noncommuting tests.

1.9 Expectation Value Means Average Value

The standard term *expectation value* can be misleading. Imagine a position probability density concentrated in two symmetric, separated lobes. Its average position is zero, even if the probability of finding the particle near zero is tiny or vanishes. The average need not be a likely single outcome.

Let $\hat{K}$ be a Hermitian observable with a complete orthonormal eigenbasis,

$$\hat{K} |n\rangle = \lambda_n |n\rangle, \quad \sum_n |n\rangle \langle n| = \mathbf{1}. \quad (1.42)$$

For a normalized state $|\psi\rangle$, define

$$\langle \hat{K} \rangle_\psi := \langle \psi | \hat{K} | \psi \rangle. \quad (1.43)$$

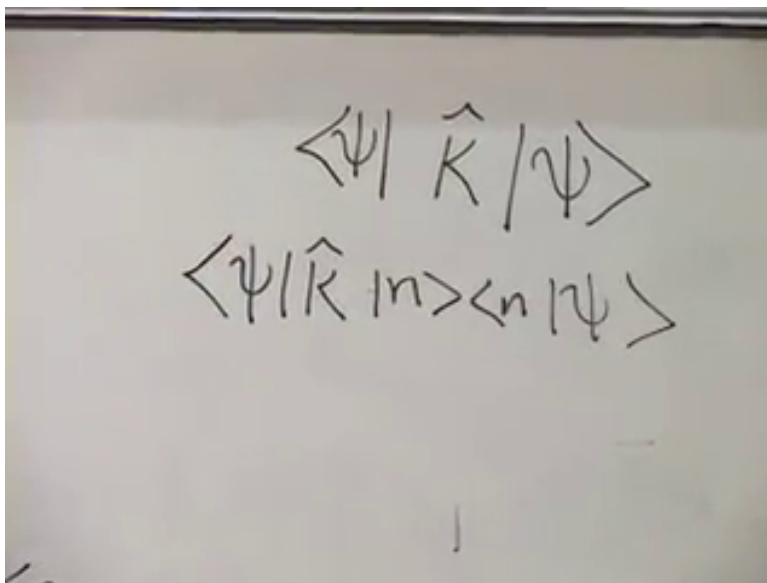


Figure 1.14: The expectation-value sandwich after inserting the eigenbasis resolution of identity.

Lecture frame: 01:39:45.

Question from the audience. Should there be a complex-conjugation star on $\hat{K}$?^a

Response. No additional star is needed. Turning $|\psi\rangle$ into $\langle\psi|$ already takes the complex conjugate transpose. The observable itself is Hermitian, $\hat{K}^\dagger = \hat{K}$, which ensures that the expectation value is real.

^aLecture timestamp: 01:38:17–01:38:41.

Insert the eigenbasis resolution of identity between $\hat{K}$ and $|\psi\rangle$:

$$\langle \hat{K} \rangle_\psi = \langle \psi | \hat{K} \left(\sum_n |n\rangle \langle n| \right) | \psi \rangle \quad (1.44)$$

$$= \sum_n \langle \psi | \hat{K} | n \rangle \langle n | \psi \rangle \quad (1.45)$$

$$= \sum_n \lambda_n \langle \psi | n \rangle \langle n | \psi \rangle \quad (1.46)$$

$$= \sum_n \lambda_n |\langle n | \psi \rangle|^2. \quad (1.47)$$

The factor

$$p_n = |\langle n | \psi \rangle|^2 \quad (1.48)$$

is the probability of obtaining λ_n . Thus $\langle \hat{K} \rangle_\psi = \sum_n p_n \lambda_n$ is exactly an ordinary weighted average of the possible outcomes.

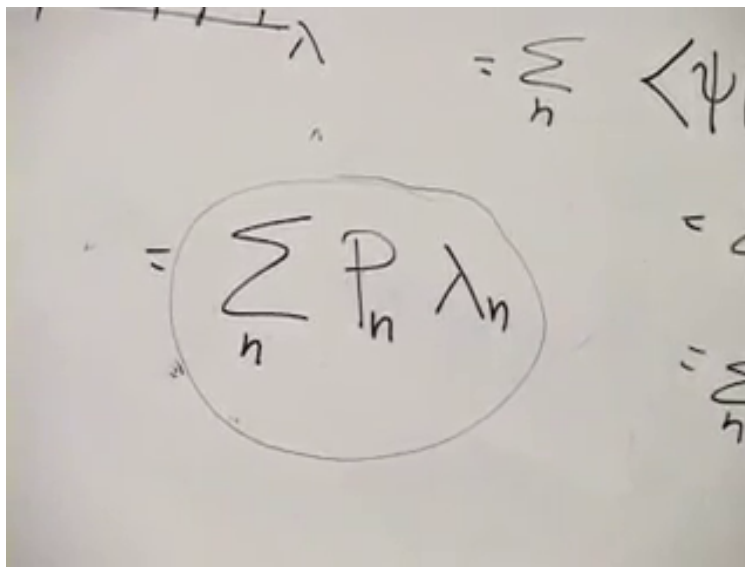


Figure 1.15: Possible eigenvalues weighted by their probabilities give the average $\sum_n p_n \lambda_n$.

Lecture frame: 01:42:50.

Return finally to the axial polarization observable in the state $|\theta\rangle$:

$$\langle \hat{P}_0 \rangle_\theta = \langle \theta | \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} | \theta \rangle \quad (1.49)$$

$$= \begin{pmatrix} \cos \theta & \sin \theta \end{pmatrix} \begin{pmatrix} \cos \theta \\ -\sin \theta \end{pmatrix} \quad (1.50)$$

$$= \cos^2 \theta - \sin^2 \theta = \cos 2\theta. \quad (1.51)$$

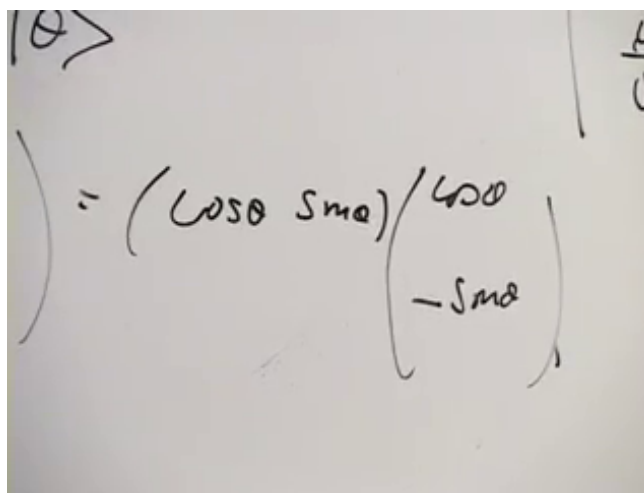


Figure 1.16: The axial polarization expectation value in the state $|\theta\rangle$.

Lecture frame: 01:45:50.

The same answer follows directly from the two outcomes:

$$(+1) \cos^2 \theta + (-1) \sin^2 \theta = \cos 2\theta. \quad (1.52)$$

No single detector click reports $\cos 2\theta$. Each run still records $+1$ or -1 ; $\cos 2\theta$ emerges only as the average over repeated identically prepared photons. That distinction between a state, a single-shot outcome, a probability, and an ensemble average completes the two-state polarization example.